



Project Title - Seasonal Agriculture Performance Analysis

NAME: VIKASH K

**COLLEGE NAME:Shanmugha Arts,
Science, Technology & Research
Academy**

AICTE STU

ID:STU6a69a01e033cc1785307166

Vikash K



PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another.
- However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed under different seasonal conditions.
- The project aims to analyze the agricultural dataset and identify meaningful seasonal patterns, trends, relationships and variations in agricultural performance.



Project Description

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- This project analyzes agricultural data collected across different seasons, crops, states and farming conditions.
 - The dataset contains 4,000 agricultural records and 28 attributes.
 - The analysis covers environmental conditions such as rainfall, temperature, humidity and soil moisture.
 - Farming practices such as irrigation, fertilizer, pesticide usage and seed quality are also analyzed.
 - Agricultural performance is evaluated using yield, production, revenue, cost and profit.
 - Resource utilization is studied using water usage and water efficiency.
 - Statistical and visualization techniques are used to identify relationships, seasonal differences and unusual patterns.
 - The final analysis provides data-driven insights for better seasonal agricultural planning.
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WHO ARE THE END USERS?

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1. Farmers
 - Compare crop performance across seasons
 - Understand resource requirements
 - Support crop and irrigation planning
 2. Agricultural Planners
 - Identify seasonal trends
 - Support agricultural planning and resource allocation
 3. Government & Agricultural Departments
 - Understand regional agricultural performance
 - Support evidence-based agricultural policies
 4. Agricultural Researchers
 - Study relationships between environmental conditions and crop performance
 - Identify unusual seasonal patterns
 5. Data Analysts
 - Analyze agricultural trends
 - Develop data-driven insights and visualizations
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Technology Used

Technology	Purpose
Python	Data analysis and processing
Google Colab	Development environment
Pandas	Data cleaning and manipulation
NumPy	Numerical analysis
Matplotlib	Data visualization
Seaborn	Statistical visualization
SciPy	Statistical analysis
CSV Dataset	Source agricultural data
GitHub	Project/code repository



RESULTS

```
-----
Total Records      : 4000
Total Attributes    : 28
Crops               : 8
States             : 8
Districts          : 10
Seasons            : 3
Duplicate Records   : 0
Missing Values      : 120
```

2. SEASON DISTRIBUTION

```
-----
Season
Kharif      1779
Rabi        1627
Zaid         594
Name: count, dtype: int64
```

3. SEASONAL PERFORMANCE

```
-----
```

	Yield	Production	Revenue	Cost	Profit	Water_Used \
Season						
Kharif	5.64	46.31	710719.06	531804.41	178914.65	6102.20
Rabi	5.08	41.49	601526.05	513836.58	87689.47	5846.99
Zaid	4.67	38.89	519171.90	543976.73	-24804.82	6419.89



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RESULTS

Kharif	5.89	54.47
Rabi	5.19	40.48
Zaid	4.41	38.22

4. KEY FINDINGS

Highest Average Yield : Kharif (5.64 tonnes/ha)
Lowest Average Yield : Zaid (4.67 tonnes/ha)
Highest Average Profit : Kharif (₹ 178914.65)
Highest Water Usage : Zaid (6419.89 m³)
Highest Water Efficiency : Kharif (5.89)

5. TOP CORRELATIONS WITH YIELD

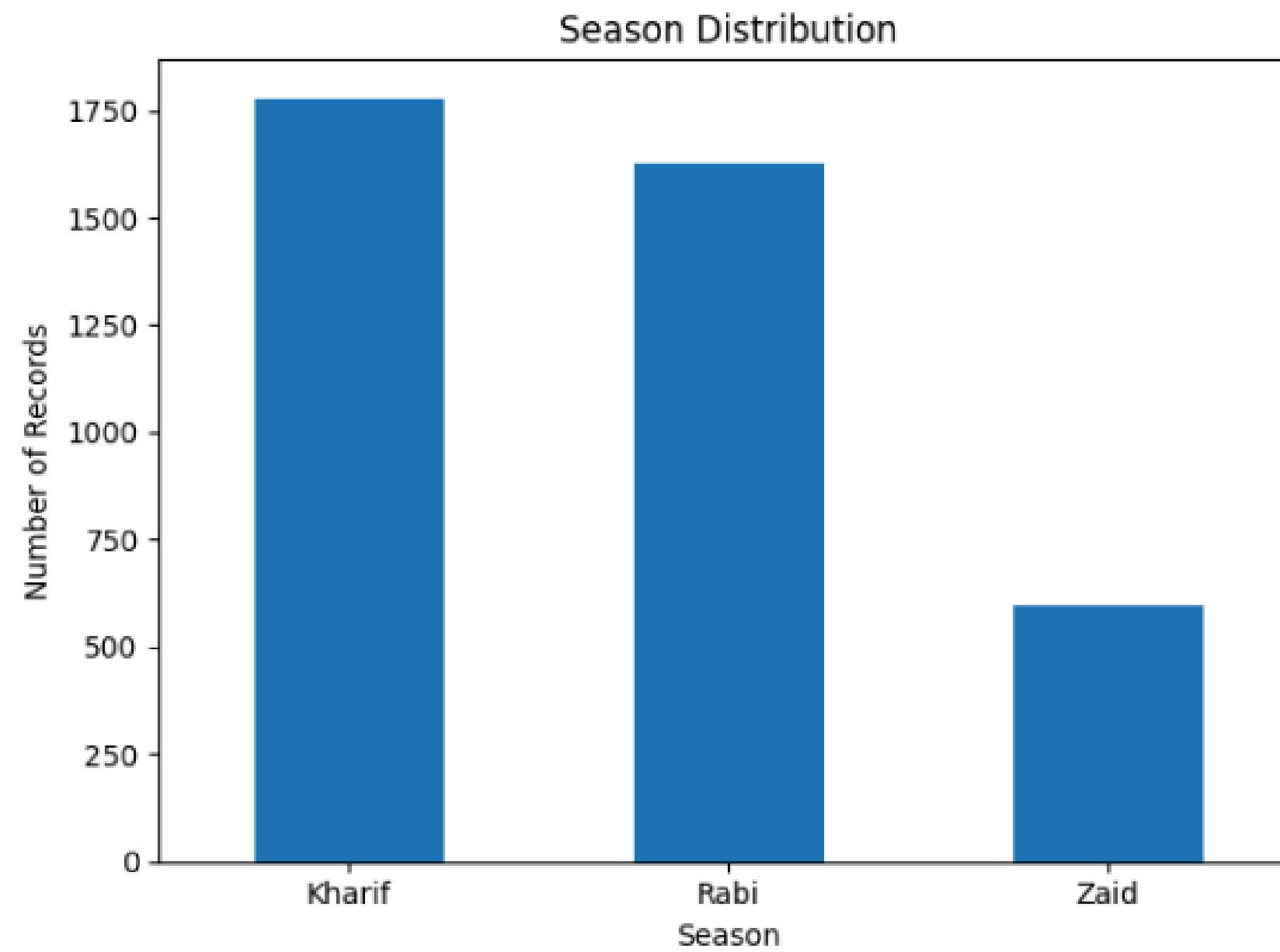
Water_Efficiency_t_per_1000m3 0.915
Production_Tonnes 0.885
Profit_INR 0.490
Revenue_INR 0.434
Water_Used_m3 0.389
Name: Yield_Tonnes_Ha, dtype: float64

6. ANOVA - YIELD ACROSS SEASONS

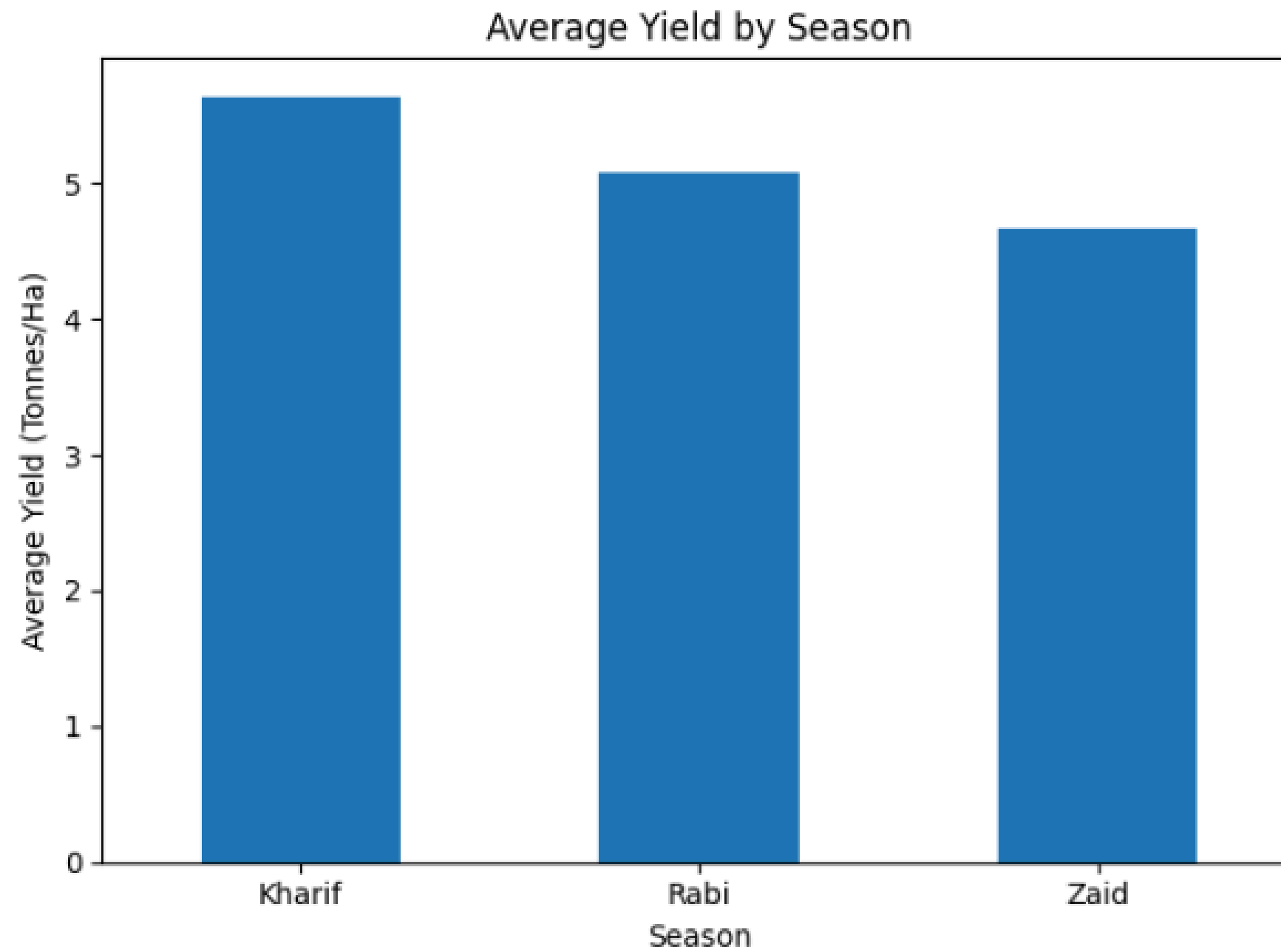
F-statistic : 1.458
p-value : 0.233
Conclusion : No statistically significant difference

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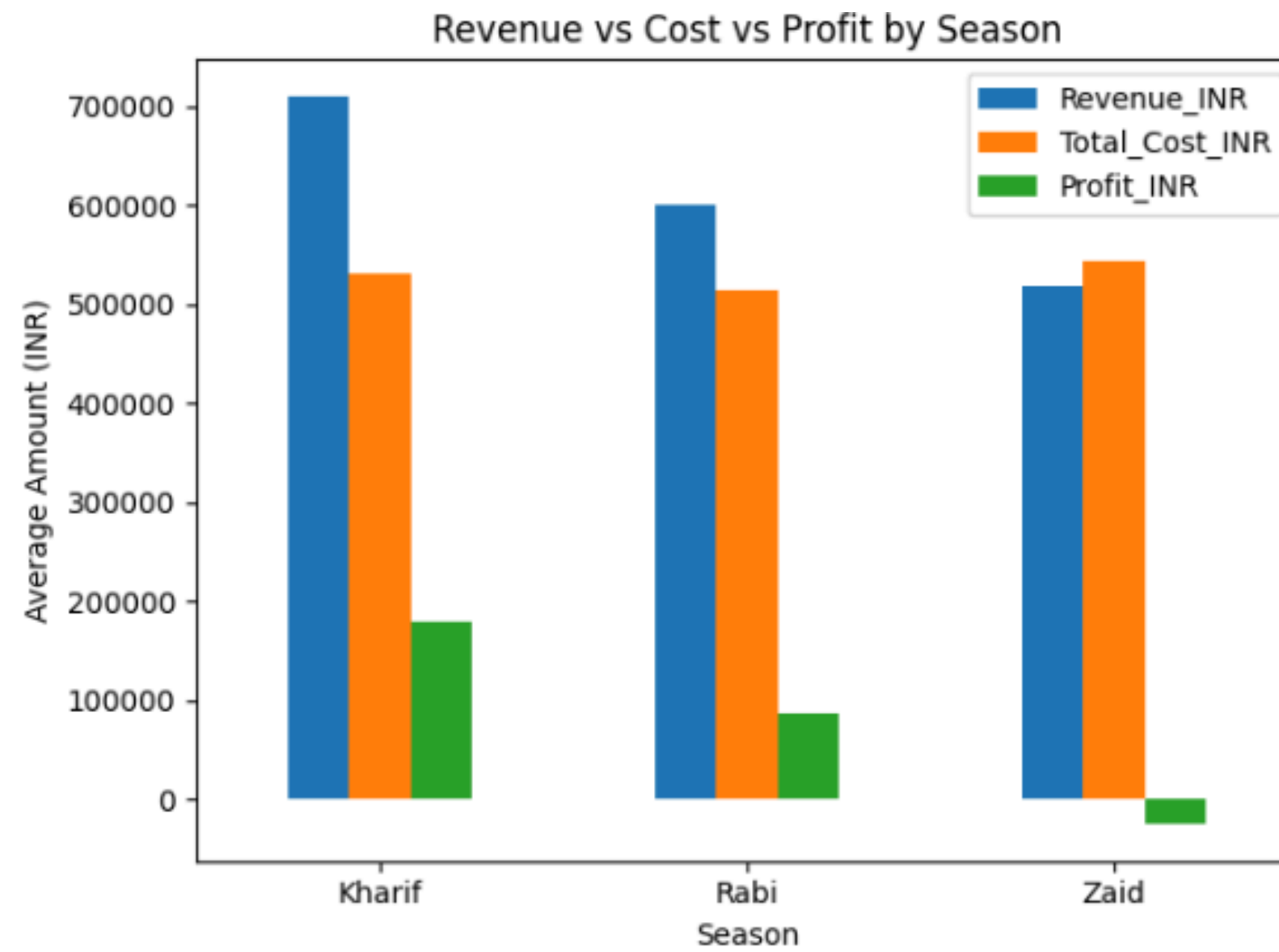
RESULTS



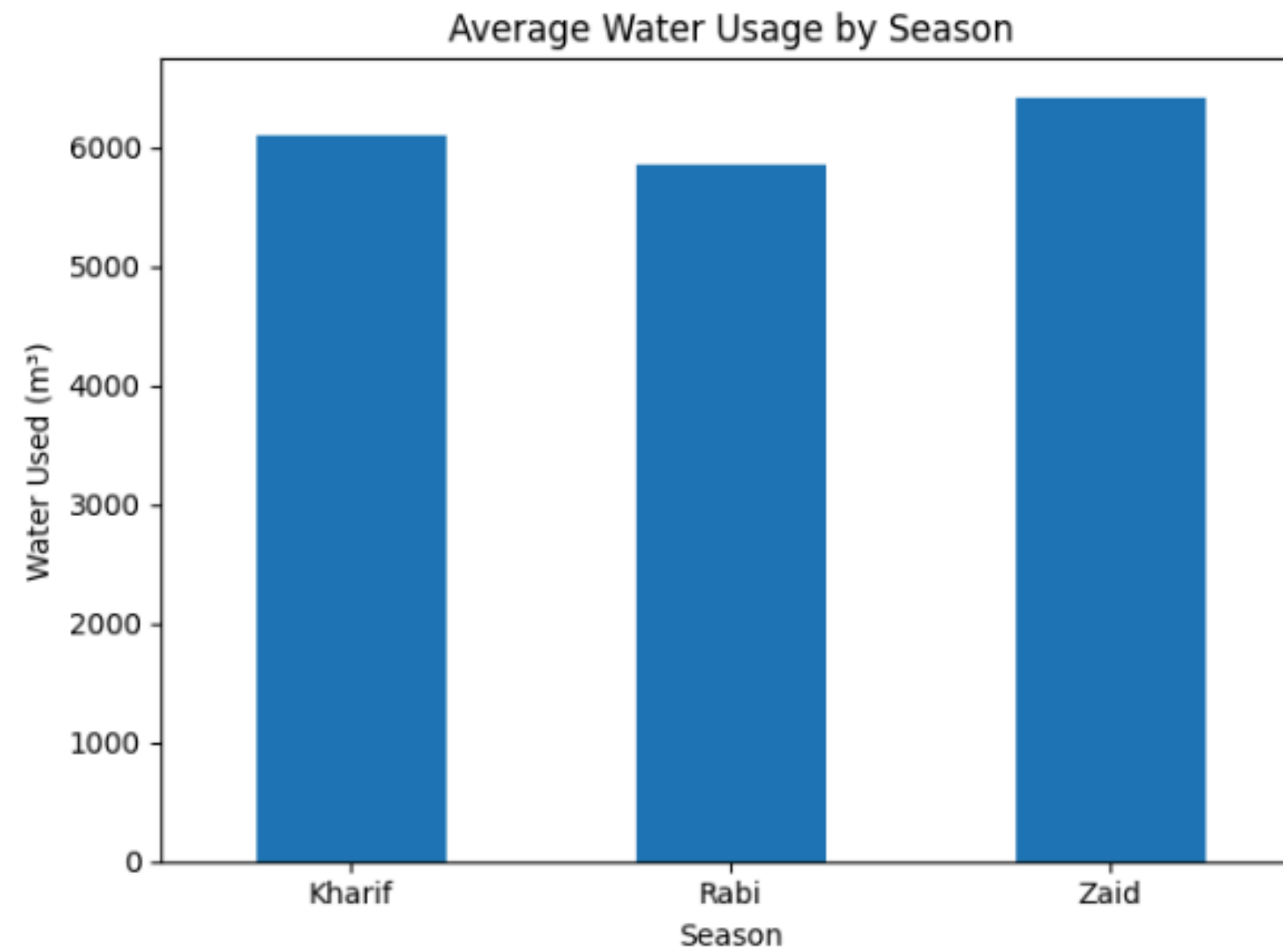
Result



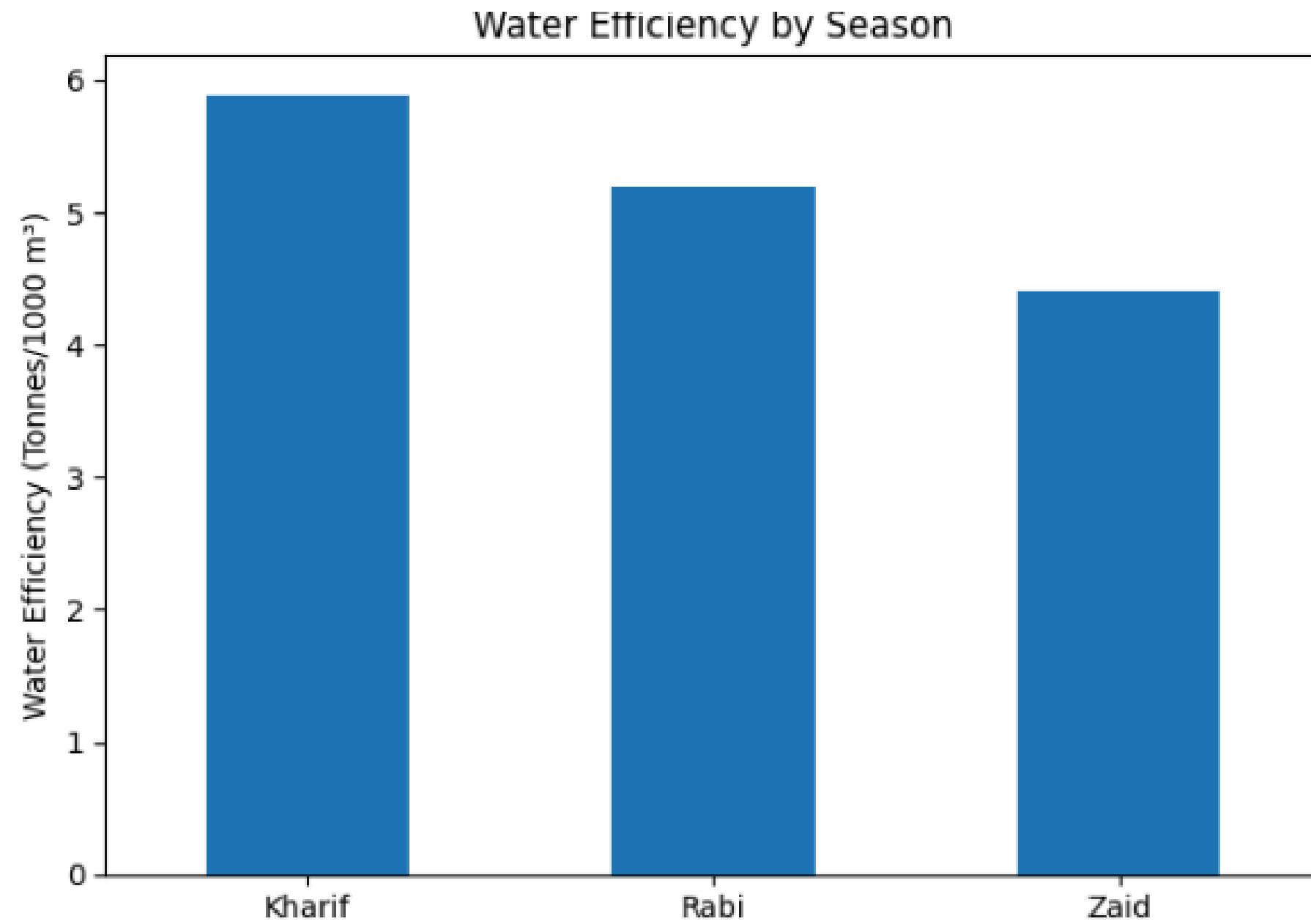
Result



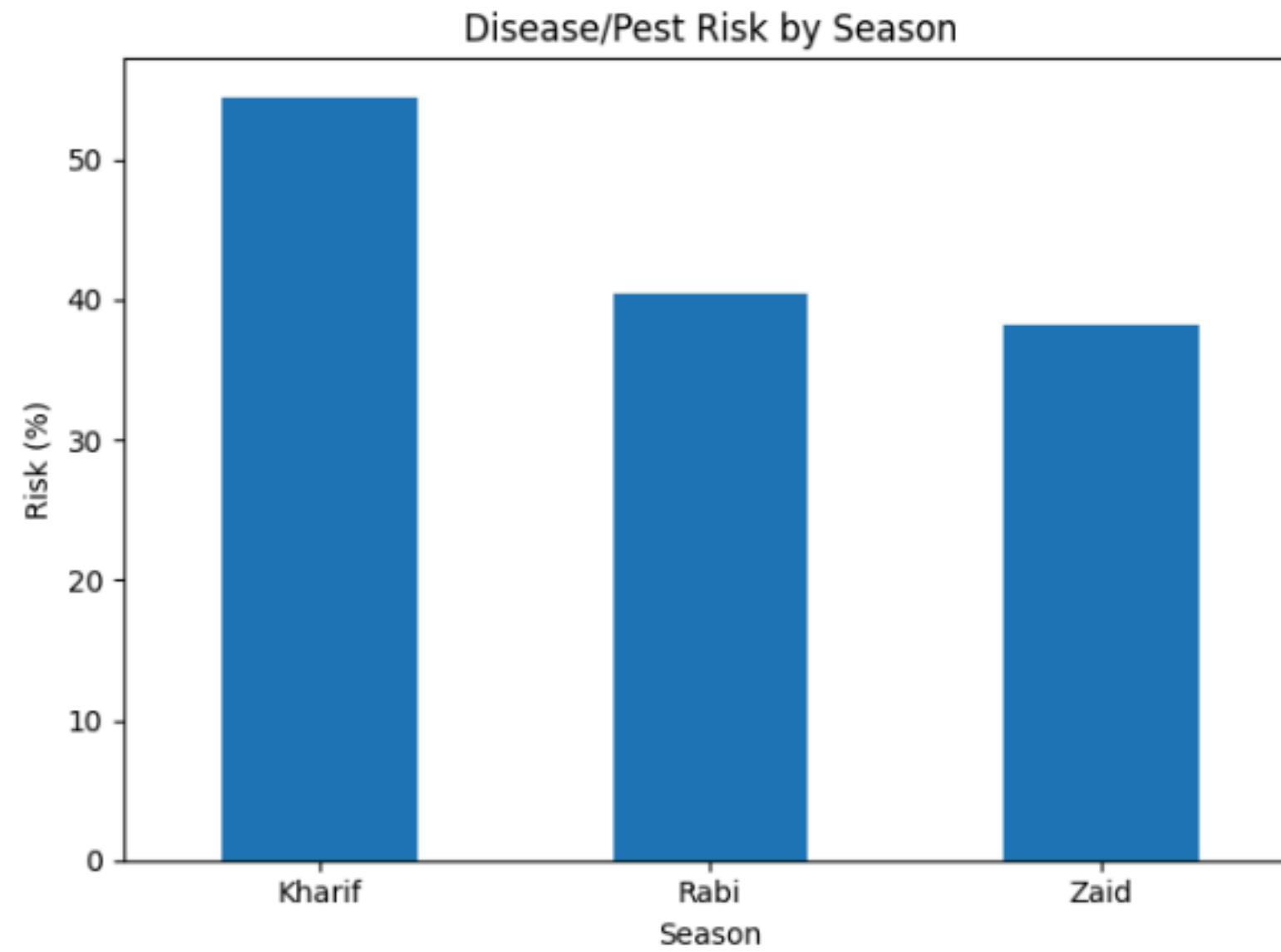
Result



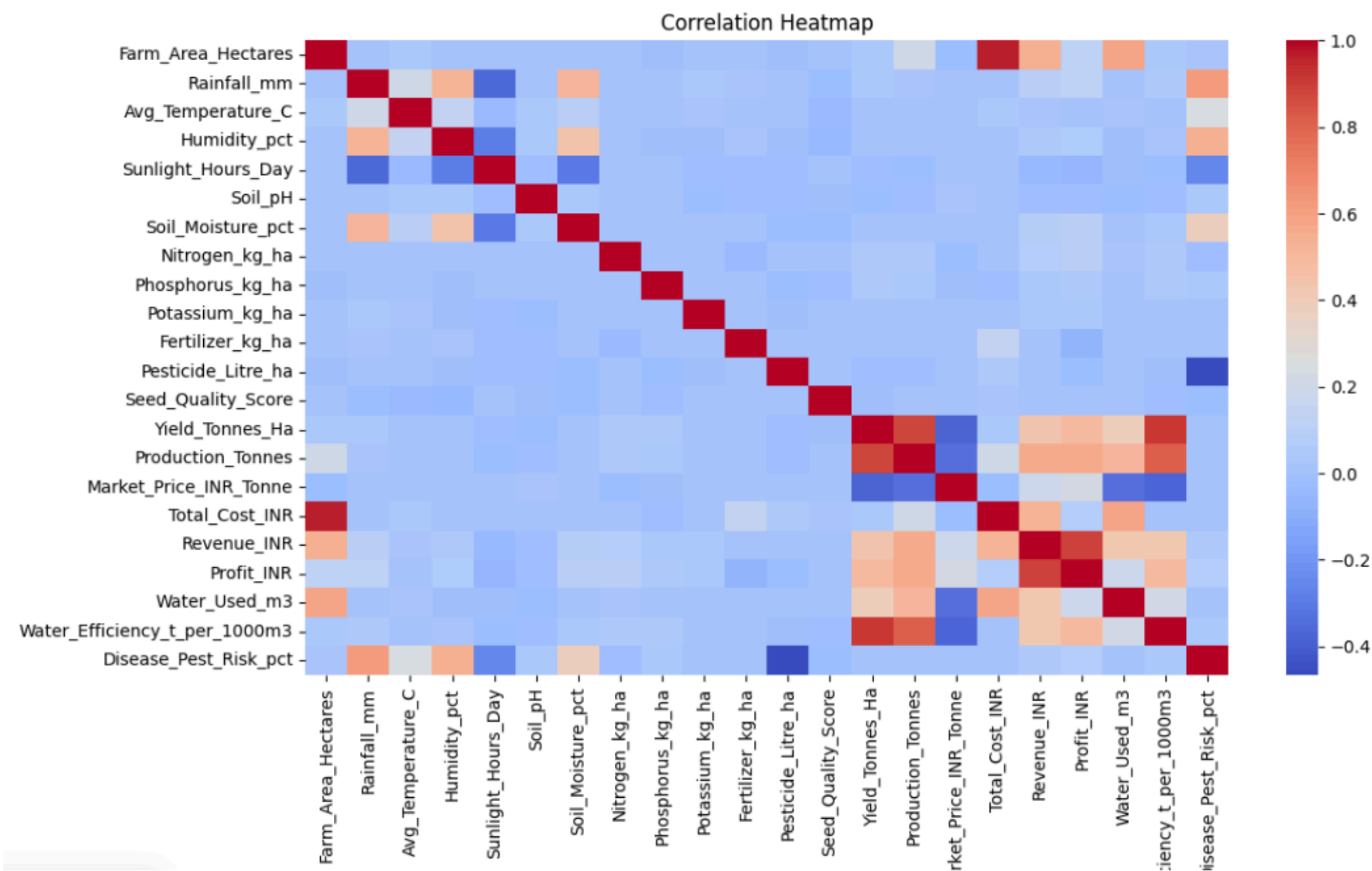
Result



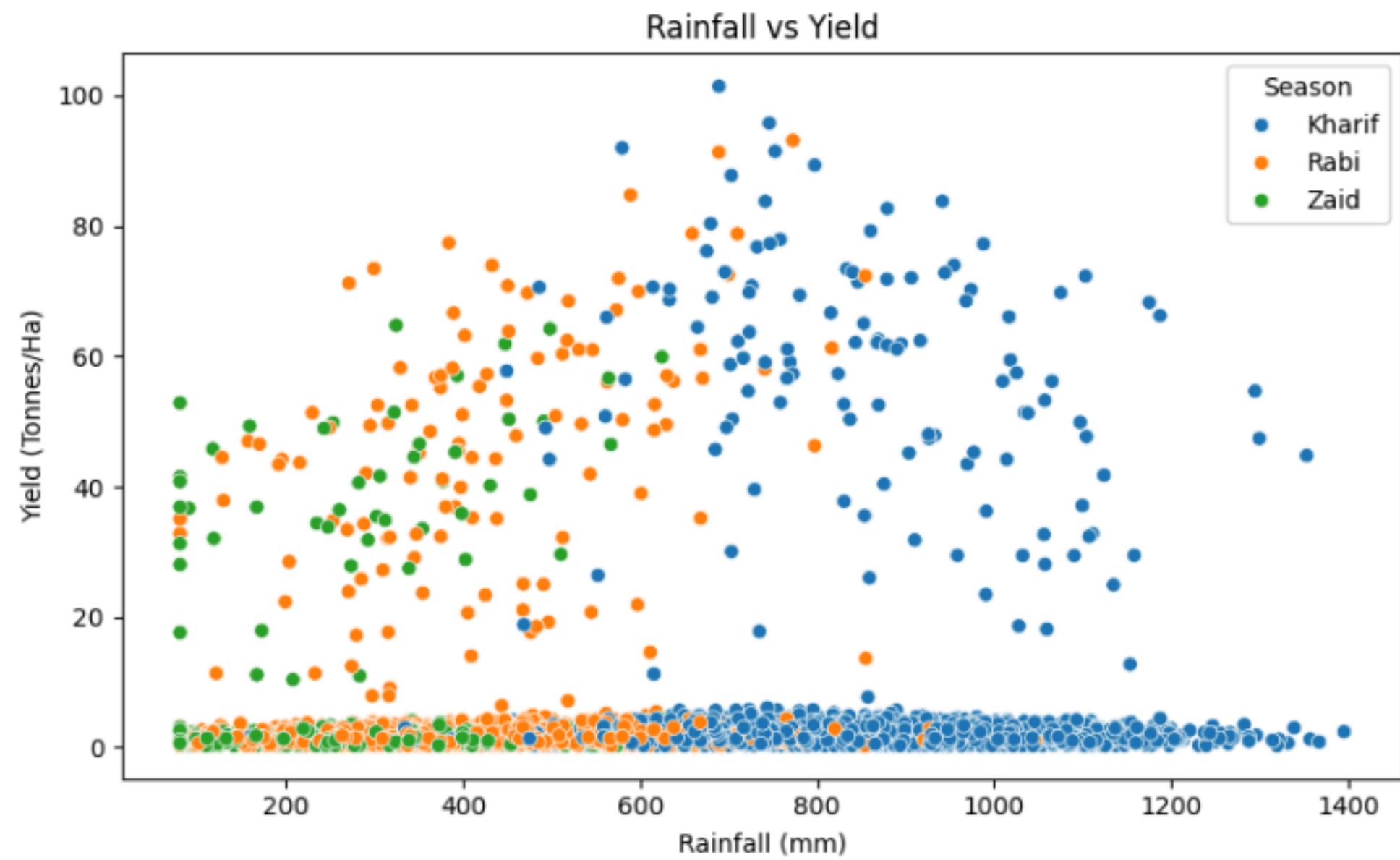
Result



Result



Result



Future scope

- Yield Prediction: Develop machine learning models to predict crop yield using weather, soil, irrigation, and farming data.
- Crop Recommendation: Recommend the most suitable crop based on soil conditions, season, rainfall, and expected profitability.
- Smart Irrigation: Use water-usage and efficiency data to optimize irrigation and reduce water wastage.
- Weather Integration: Integrate real-time weather data for better seasonal planning.
- Disease & Pest Prediction: Build models to identify and predict disease and pest risks.
- Profit Forecasting: Predict future revenue, production costs, and profitability for different crops and seasons.
- Interactive Dashboard: Develop a real-time dashboard for farmers and agricultural planners.
- Regional Analysis: Expand the analysis to more states, districts, crops, and geographical regions.
- Real-Time Market Analysis: Integrate market-price data to support better crop selling and planning



GitHub Link

<https://github.com/AswiniKumar55/-Seasonal-Agriculture-Performance-Analysis-.git>



VOIS Course completion certificate - Course name- Data Visualization



Thank you

